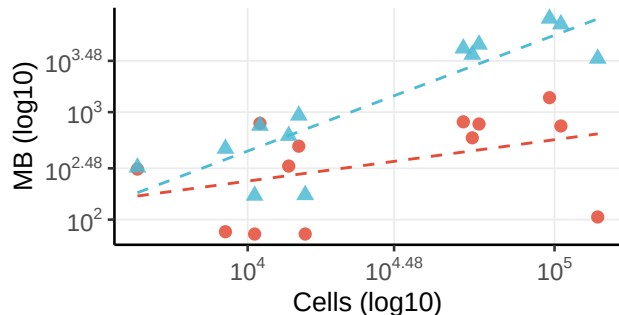
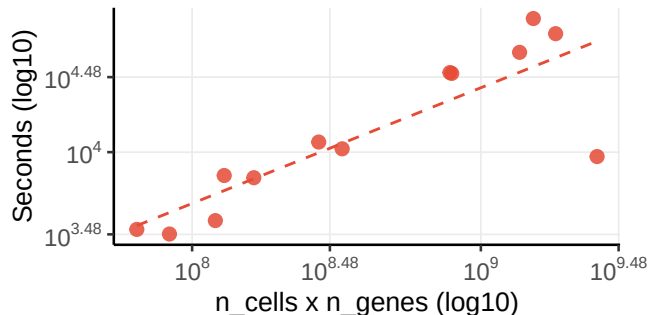
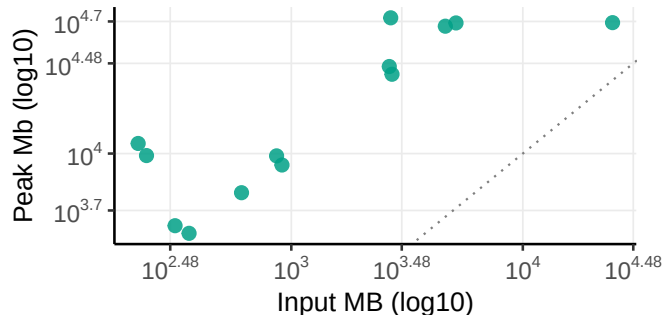
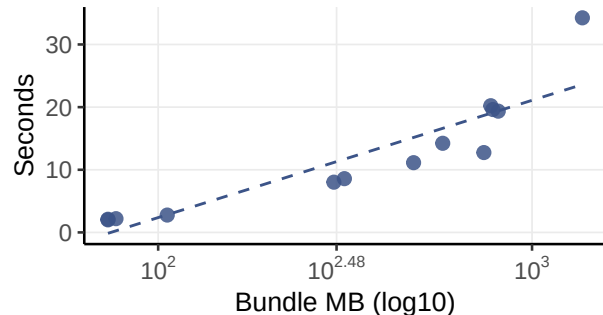
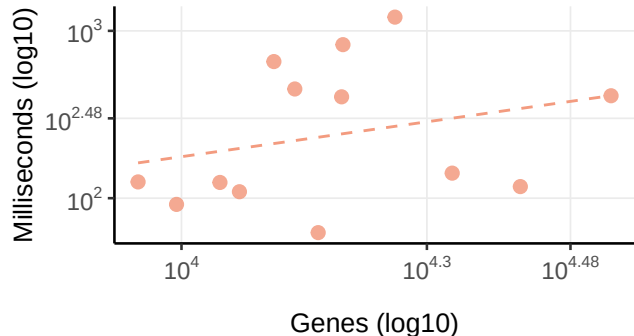


A**Bundle + shard tree size vs cell count**

● Bundle (.scminer.h5) ▲ Shard tree (.csv.gz)

**B****prepare_study() wall time****C****Peak memory vs total input size****D****load_study() cold-start latency****E****gene_values() fetch latency (median)****F****Output : input size ratio per study**